Fiscal Year:	FY 2009	**Task Last Updated:**	FY 05/30/2009
PI Name:	Green, Lora Ph.D.		
Project Title:	Mechanism of Radiation-Induced Alteration of Astrocyte-Neuronal Coupling		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92354	**Congressional District:**	41
Comments:	PI is deceased as of January 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	10/01/2006	**End Date:**	09/30/2009
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nelson, Gregory (Loma Linda University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	The goal of the proposed project addresses the underlying mechanistic basis for radiation-induced alteration in the coupling between astrocytes and neurons. Our goals correspond to one of NASA's critical questions, specifically to "Acute and Late CNS Risks" from the critical path roadmap (risk 29). Objective: To better understand the mechanistic changes occurring in the astrocytes-neuron couple following exposure to radiation. We hypothesize that ionizing radiation disrupts the metabolic coupling between neurons and astrocytes by altering glutamate transport and metabolism. We propose to investigate the mechanism for this uncoupling by measuring alterations in glutamate uptake and metabolism, and to determine how abnormal glutamate uptake affects intercellular signaling following irradiation. To achieve these objectives we will first utilize NTera2/D1 cells that can be induced to differentiate into mature neurons and astrocytes, and secondly confirm our finding using primary astrocytes and neurons. These differentiated cell types will be used to investigate the mechanism of neurometabolic uncoupling following acute exposure to radiation. The		

Task Description:	objectives will be achieved by the following specific aims. Aim 1: Measure glutamate transport and metabolism in human astrocytes and neuronal cultures following acute exposure to 290 MeV carbon and 1GeV iron ions delivered as single fractions of 0, 10, 50 and 200 cGy. Samples will be harvested at 3 hrs, 2 and 7 days post irradiation. This aim will also explore the physical and soluble dependency of homeostatic cooperativity between astrocytes and neurons. Aim 2: Measure the endogenous and temporal changes in expression of antioxidants and reactive oxygen species in astrocytes and neurons using the same radiation conditions described in aim 1. Aim 3: Examine alterations in inter-astrocytic and neuronal communication after exposure to ionizing radiation by analyzing essential elements of physical coupling and calcium signaling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Two runs of carbon (spring 2007 and 2008) and two runs of iron (fall 07 and fall 08) were conducted at the NSRL to collect data to complete the aims proposed in the research grant provided by NASA negotiated to be funded through the cooperative agreement given to LLU proton facility. Since the prior report the gamma radiation study of radiation-induced modulation of glutamate function has been published (Radiation Research 2009, 171:41-52). The manuscript detailing the high LET glutamate response is nearly ready to submit to radiation research. Collectively the glutamate data from the carbon and iron exposures showed similar results as the gamma ray experiments with the astrocytes having decreased glutamate uptake and the neurons having an increase in glutamate uptake. For gamma rays the response of the astrocytes was only significant at 3hrs post exposure, by the 2 day time point the cells returned to normal with respect to their glutamate uptake. The neurons which normally do not take up glutamate show a radiation induced increase that stays elevated throughout the 7 day time point, the last harvest point in these experiments. The results of the sustained glutamate uptake in the neurons showed some dose dependence with higher doses persisting through the 7 day point post irradiation. The difference with respect to high LET radiation was that the duration of the change in the astrocytes persisted longer, with depressed glutamate uptake through the 2 day time point. The neurons response to LET was reduced magnitude of the increased glutamate uptake. In general the magnitude of the signal was inversely proportional to the LET. In addition to conducting experiments with the isolated astrocytes and neurons we also conducted sets wherein the astrocytes and neurons were mixed at a 10 to 1 ratio and allowed to establish for 4 days prior to exposure to radiation. The results for glutamate function and apoptosis following gamma rays, protons, carbon, and iron ions were muted with respect to the same measurement taken from the isolated cell types. The oxidant/antioxidant levels measured for gamma rays were also muted in the mixed cultures (described below). The LET effect on radiation-induced changes in glutamate uptake lead to the possibility that the change may be due to fluence differences which decreases as the LET increases. The other possibility is the production of reactive oxygen species which also decrease as the LET increases. The first measurements done in attempt to explain changes in function were to measure the levels of the transporters expressed on the respective cell types. The changes in glutamate uptake were not due to an overt change in the levels of the transporter' protein levels as western blots and quantitative immunocytochemistry confirmed that the transporters did fluctuate with time but not with dose of radiation. The change in function was also not fully attributed to reactive oxygen species as we mimicked the levels using paraquot to stimulate super oxide levels and only saw a very slight change in the glutamate uptake. This idea should be further explored to rule out ROS as a suspect in the radiation-induced changes in glutamate uptake measured in these experiments. In that regard in completing aim 2 we did significant investigation of oxidant/antioxidant activity following exposure to gamma radiation. A complete set of oxidant/antioxidant have been conducted following carbon and iron exposure and the samples stored frozen. We are now starting the assays these samples to complete that part of aim 2.
Bibliography Type:	Description: (Last Updated: 01/29/2015)
Abstracts for Journals and Proceedings	Sanchez MC, Bianski B.M, Ortloff LS, Green L.M. "Glutamate Uptake Effects after Irradiation with Iron and Carbon Ions." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30th -July 2nd, 2008. 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30th -July 2nd, 2008. , Jul-2008
Abstracts for Journals and Proceedings	Sanchez MC, Bianski BM, Ortloff LS, Green LM. "Effect of Iron (1GeV 56Fe) on Glutamate Transport in Astrocytes, Neurons and Mixed Cultures." Radiation Research Society Meeting, Boston, MA, September 21-24, 2008. Radiation Research Society Meeting, Boston, MA, September 21-24, 2008. , Sep-2008
Abstracts for Journals and Proceedings	Sanchez MC, Bianski BM, Ortloff LS, Green LM. "The Effect of Protons and Particle Radiation on Glutamate Transport in Astrocytes, Neurons, and Mixed Cultures." 40th Annual Meeting of the ASN, Charleston, South Carolina, March 7-11, 2009. 40th Annual Meeting of the ASN, Charleston, South Carolina, March 7-11, 2009. , Mar-2009
Articles in Peer-reviewed Journals	Sanchez MC, Bianski BM, Ortloff LS, Green LM. "The Effect of Protons and Particle Radiation on Glutamate Transport in Astrocytes, Neurons, and Mixed Cultures." Radiation Research , submitted, May 2009. , May-2009
Articles in Peer-reviewed Journals	Sanchez MC, Benitez A, Ortloff L, Green LM. "Alterations in glutamate uptake in NT2-derived neurons and astrocytes after exposure to gamma radiation." Radiat Res. 2009 Jan;171(1):41-52. PubMed PMID: 19138048 , Jan-2009